

./03-database/DOC-05-Database-Design.md

DOC-05: Database Design Document

Version: 1.0 | **Status:** APPROVED | **Date:** 2026-08-27 **Purpose:** Conceptual and logical database architecture.
Owner: Database Architect

Database Principles

- **Naming Convention:** snake_case + plural for tables (e.g., organizations, users).
- **Primary Keys:** UUID id for all tables.
- **Timestamps:** created_at, updated_at, deleted_at (nullable).
- **Soft Delete:** Evaluated entity-by-entity (e.g., users, organizations), not blindly applied.
- **Tenant Isolation:** organization_id in all tenant-owned tables. Composite indexes on (organization_id, created_at).

Core Entities

1. **organizations:** id, name, slug, status.
2. **users:** id, email, password_hash, status.
3. **organization_members:** id, organization_id, user_id, status.
4. **roles & permissions:** roles, permissions, role_permissions, member_roles.
5. **audit_logs:** id, organization_id, actor_user_id, action, resource, metadata.

```
erDiagram
    ORGANIZATIONS ||--o{ ORGANIZATION_MEMBERS : has
    USERS ||--o{ ORGANIZATION_MEMBERS : belongs_to
    ORGANIZATION_MEMBERS ||--o{ MEMBER_ROLES : assigned
    ROLES ||--o{ MEMBER_ROLES : contains
    ROLES ||--o{ ROLE_PERMISSIONS : has
    PERMISSIONS ||--o{ ROLE_PERMISSIONS : granted
```